

Current Sensing – UPS Downtime Detection

Course Project: CCIoT (EC5.204)
Team: SSRH

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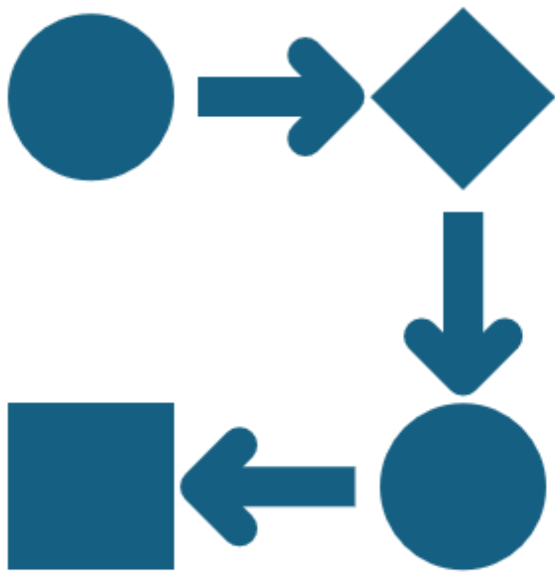
Table of Contents

- **Original Problem Statement**
- **Changes in Problem Statement**
- **System Identification**
 - Visualizing the problem statement
- **Components**
- **Communication and Dashboard**
 - Wi-Fi connection
 - Blynk
- **Deployment**
- **Journey**
 - Iteration 1
 - Iteration 2
 - Iteration 3
- **Insights**
- **Learnings**
- **Scope of the project**
- **Conclusion**

Problem Statement

- The main aim of this project was to **detect and notify** the **downtime of the UPS** based on the input and output current of the machine.
- Downtime is a very rare event as it generally happens either due to **bad weather conditions** or due to a **very high load on the wire** corresponding to any one of the three phases.
- Downtime can last for a **few seconds to a few hours**.

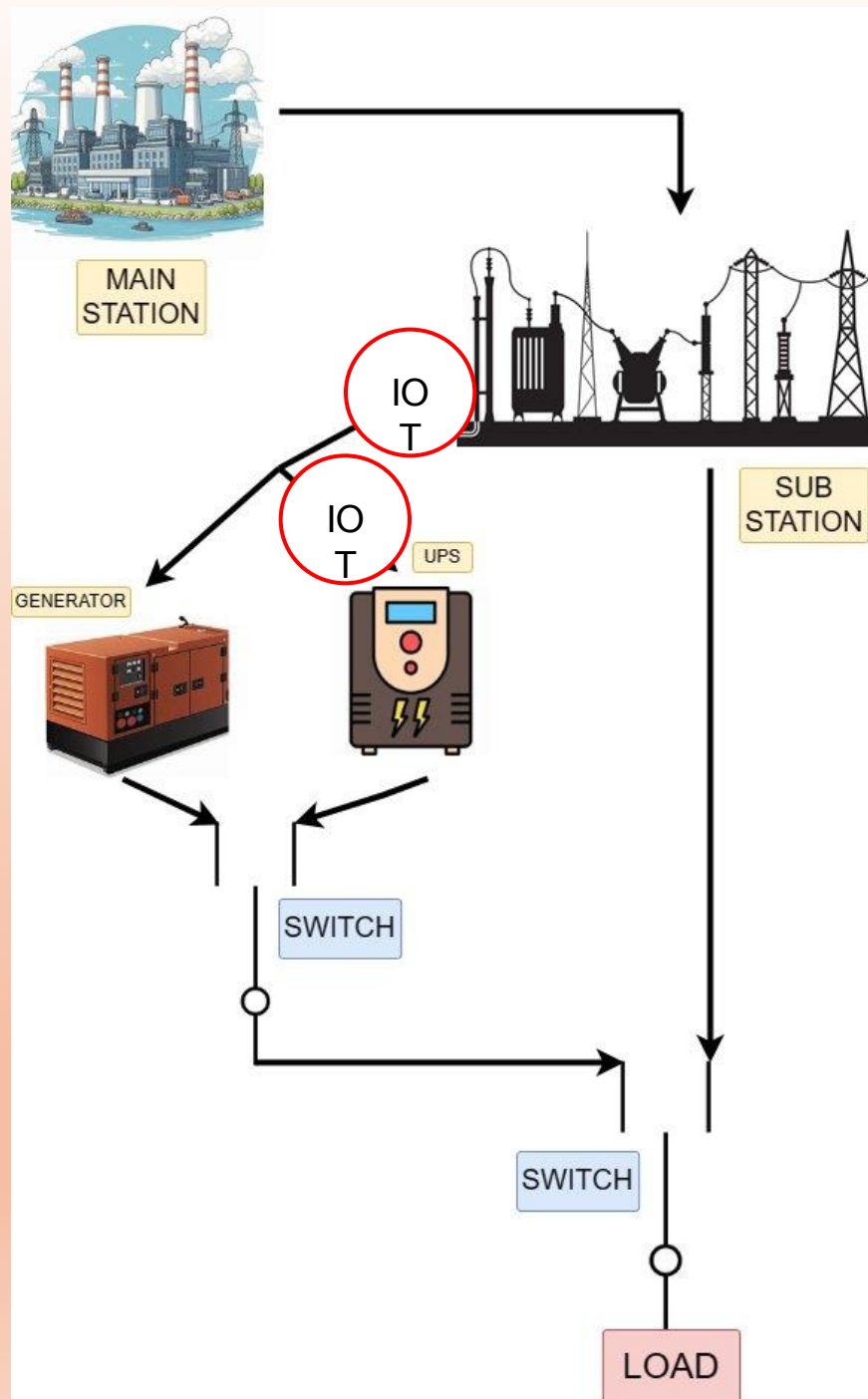




Changes in Problem Statement

- The problem statement and the requirements of the project, however, changed with time as we met more often to the client of our project.
- As opposed to the initial idea of detecting the downtime of the UPS, we were suggested to detect the **downtime of the MCB switch** that served as an intermediary between the current from mains and the input to the UPS.
- The MCB has **3 different input and output wires** for each different phase.
- It was, therefore, suggested to us to put the sensors on **any 2 of the I/O pairs**.
- The clients also suggested that we implement a **"notification" based system** where if a downtime occurs, the users should get to know about the same.

System Identification



Proposed Solution

Approach

- **Detect current drawn** through main line, sub-stations and UPS/generators
 - Use current sensors
- **Analyse discrepancies** in sensed current levels
- Use **micro-controller to relay the current data** to the communication module
- **Communicate** the current data over desired range
- Display the **data in real-time** on dashboard
- Draw **real-time analysis** - power consumption, current drawn, overload, power-cut, downtime, etc.

Components

- We have used the following components:
 - **Current transformer (0-100A) - 4:** To measure current flowing through 2 pairs of different phases
 - **Arduino Wi-fi Rev2:** As a microcontroller which has Wi-Fi modalities built-in
 - **Li-Ion Batteries:** To power the microcontroller in case of outage in the main line
 - **Battery charging module:** To charge the Li-ion batteries
 - **Battery bank:** In case the batteries drain off, we can switch to battery bank (this was an impromptu decision that we thought off which sounded more efficient than Li-ion batteries)
 - **Jio-Fi (for Wi-Fi hotspot):** because **wifi@iiith** has very weak and unrecognizable signals in the **T-Hub deployment location**

Date:
08/05/24

Communication and Dashboard

Wi-Fi connection

- In the earlier phases of implementation of the project, we had connected the **Arduino to the college Wi-Fi (wifi@iiith)**. Although we were able to connect to it, it being a Wi-Fi WPA2 Enterprise connection, gave us a **lot of issues** when the Arduino had to connect and reconnect a lot of time, primarily due to the time required for authentication.
- The problem only increased at the site of deployment, because the **Wi-Fi strength** there was **very weak**, especially inside the deployment room.
- Considering all of this, we decided to make a **switch to the Jio Wi-Fi**. Apart from being easy to connect, which helped in reducing latency, it had the added benefit on being a cellular connection, which meant that connectivity was ensured even during a power outage.

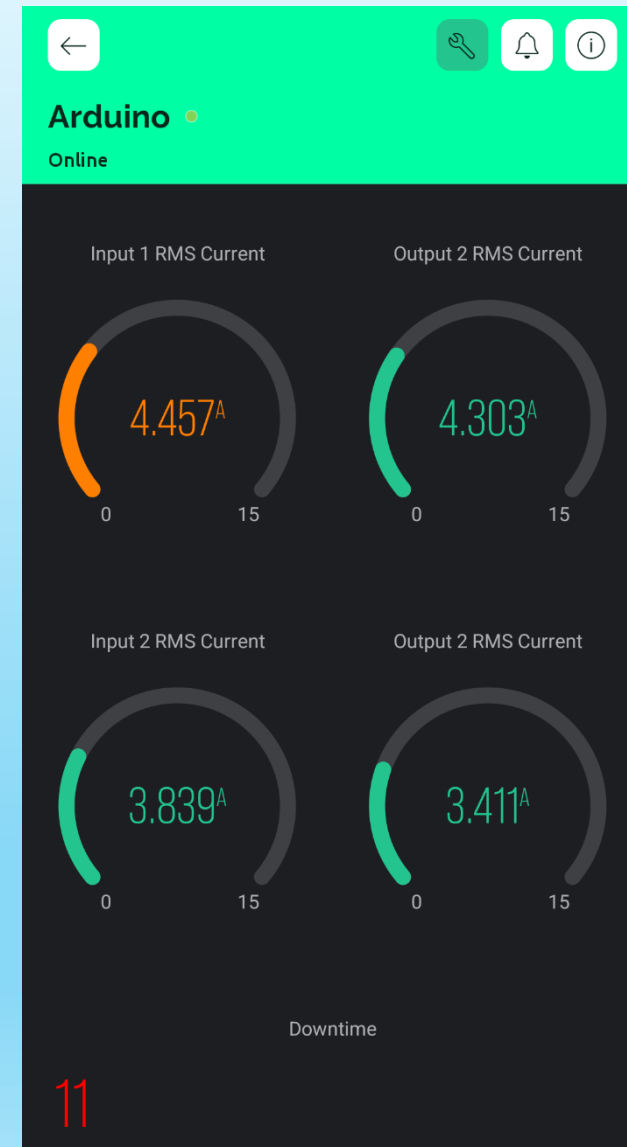
Blynk Dashboard



Blynk Dashboard

That's not all, we have designed our mobile Blynk Interface also which can be used to check the system anytime, anywhere!

It also alerts the user of any downtime instantly!



Blynk Notification System

× Notifications & Events [Notifications Settings](#)

1d 1w 1mo ● ●

📶 All 1 Critical Warning 1 Info Content Resolved

- Online 11:03:17 PM Yesterday

Offline for 28s

- Offline 11:02:49 PM Yesterday
- **Downtime Detected** 9:15:05 PM Yesterday
Time Since Last Downtime: 1791ms, Reason: ...
- Online 6:10:48 PM Yesterday

Notification received on Blynk Dashboard

Arduino: Downtime Detected [Inbox ×](#)

 **Blynk** <robot@blynk.cloud>
to me ▼

Downtime Detected

Time Since Last Downtime: 4366ms, Reason: Power Failure

[Open in the app](#) | [Mute notifications](#)

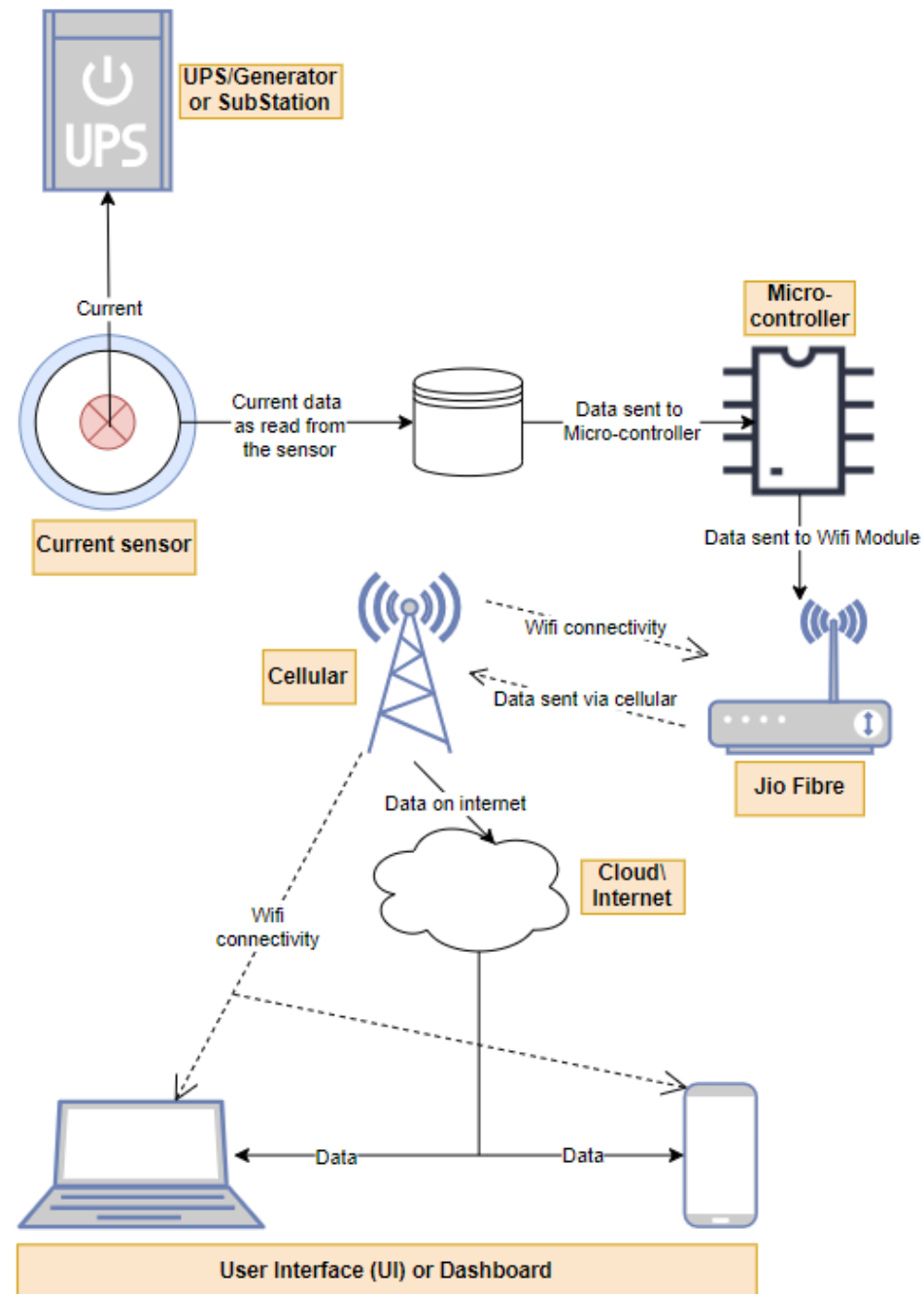
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Date: Tuesday, May 7, 2024, 4:11:28 AM India Standard Time
Device name: [Arduino](#)
Organization: [My organization - 5092GW](#)
Template: Test
Owner: [soumilgupta210904@gmail.com](#)

Notification received on e-mail

Deployment

This is the set up that we deploy at the **input of the UPS** (which is the **output of the MCB**) and at the **input of the MCB** (which is the **main power line**)



Date: 08/05/24

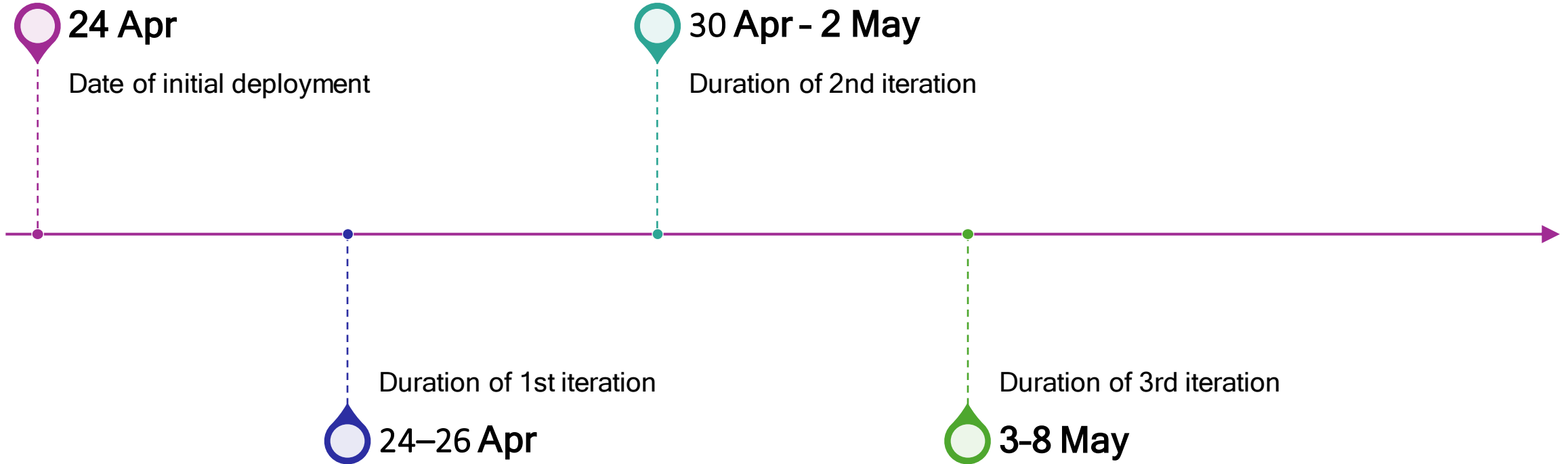
Deployment and Working

Video :

Deployment, Working
and Analysis video



Journey



Iteration 1

Date:
08/05/24

Duration of deployment: 24th - 26th April

Components deployed: 2 current sensors, 2 batteries along with battery charging modules, Jio Wi-Fi

Number of downtimes : 4

Problems encountered: The batteries could not be recharged as an extension cord was unavailable for charging the batteries through the battery charging module. Hence, the batteries drained within a couple of days.

Iteration 2

Duration of deployment: 30th April - 2nd May

Components deployed: 2 current sensors, 2 batteries along with battery charging modules, Jio Wi-Fi

Number of downtimes : 0

Problems encountered : The UPS is a 3-phase system. Hence, it has 3 input wires and 3 output wires. Absence of current in any one of the 3 wires could cause a downtime. Hence, just monitoring a single input and single output wire isn't sufficient for downtime detection

Iteration 3

Date:
08/05/24

Duration of deployment: 3rd
May - 8th May

Components deployed: 4
current sensors, 2 batteries
along with battery charging
modules, Jio Wi-Fi

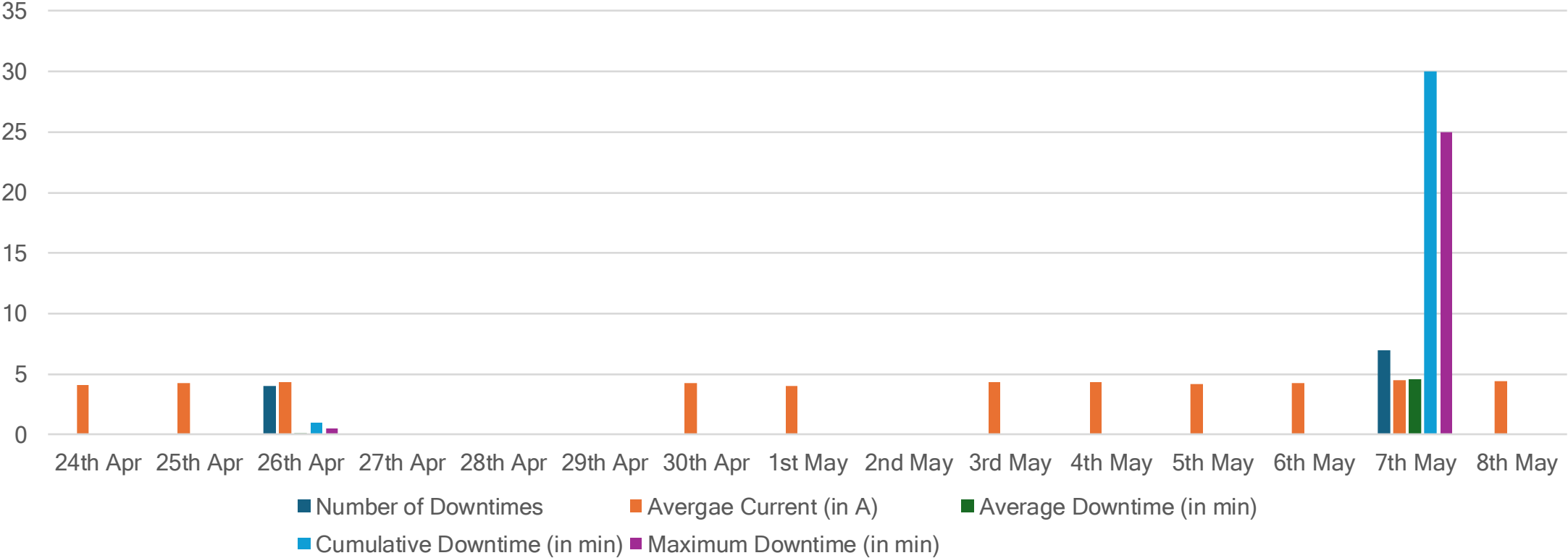
**Number of downtimes
observed:** 7

Insights

- Current across all 4 phases that we measured: **~4A**
- Downtimes:
 - Power outage from the mains = **11 times in last 11 days** (break down given later)
 - Power outage due to MCB trips = **0**
- Breakdown of downtimes
 - **4 initial minor** "under-a-minute" downtimes - not significantly captured, therefore there are chances that these may have occurred due to sudden fluctuations
 - **5th downtime** was detected on **7 May morning** at around **4.11am**
 - **6th downtime** was detected **7 May evening** at **6.09 PM** during rains and went on for around **25 minutes**
 - **7th to 11th downtimes** occurred very quickly in a span of **1 minute at 9.15 PM** and at **intervals of few seconds** (or perhaps milliseconds), again on **7 May**
 - **11th downtime** occurred at **11.03 PM** on **7 May** and went on for about **2 minutes**

Downtime Analysis

UPS Downtime Analysis



Learnings

- Learnt about **debugging IoT systems**: both hardware and software
- How to choose **appropriate sensors** for better readings
- How to **calibrate sensors**
- How to **collect and send data** over a communication protocol (and how to choose that) to the dashboard
- Producing **simpler and efficient solutions** when it comes to building the set-up and deploying the same
- How to **adapt well to the requirements of any project/client** and try to meet those requirements to the best of our knowledge
- Learnt that **deployment** is indeed the **hardest part** - in that, when it comes to maintaining and upholding the smooth functioning of the device, that's where the problem starts
- Learnt the basics of **how the college power system works**

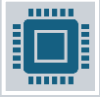
Scope of the project

- Any useful IoT application can be turned into a **billion-dollar** business when right amount of effort and ideation is put into it
- Our project is a very useful and scalable one due to following reasons:
 - **Cost effective** - sensors are not too costly. Significant cost is incurred when we use cellular communication plan which includes periodically recharging
 - **Space occupied** - the device is not significantly large so space should not be a problem
 - **Sensor calibration** - in our project, the sensor calibration part was relatively simpler
 - **Safer option** - deploying such devices is a safer option than having the power lines checked manually which involves too much risk

- The project can be scaled up for multiple use-cases as follows:
 - **Detecting power outages**
 - **Detecting** and **analyzing** power consumption
 - **Understanding the trends** in power consumption and in power outages
 - **Identification** of nodes in the power system that cause frequent issues
 - **Smart/Automatic** system that identifies power outage, and based on the downtime, makes a switch to UPS or to the generator to compensate for the outage.
 - Great industrial applications.
- There may be many other cases where this system can be used apart from the ones mentioned above.
- This is why we feel that our project has a great scope in terms of scalability

- While scalability, few of the problems that can be encountered are:
 - **Establishing connection** - jammers, interference, etc. may cause an issue.
 - **Latency** – in collection, processing, transmission and display of data.
 - **Hardware** - the wires of the current sensors are quite long, which may cause an issue while moving around
 - **Battery** - powering the microcontroller directly from the main supply will not be an efficient solution given that power outages occur most frequently in the mains. Also, longer downtimes will drain the battery (attached to microcontroller)
 - **Miscellaneous** - there may be many other peripheral issues that can come up and those will differ from situation to situation

Conclusion



The project helped us provide a sneak-peak into the world of IoT and the way it was structured gave us good insights



Making an IoT device is never easy especially when we need to be judicious in deciding on which components to choose from



Communication over a channel can get quite difficult if the location of deployment causes interferences or blockage



IoT is a trendy domain to work and explore in, and the fact that we require it now more than ever before puts emphasis on its importance



We felt delighted in making a project that had a good scope in terms of scalability and usability given that our device detects power outages without any human intervention

Acknowledgement



We would like to thank **Professor Sachin Chaudhari** for giving us an opportunity to work on such an interesting and creative CCloT project.



We would also like to thank **Prakash sir** as well as **Srinivas sir** who were in constant contact with us and provided us the access to the deployment site as well as the exact details about the UPS system.



Lastly, we would like to thank our **TA Sannidhya Gupta** for guiding us through the project and facilitating the requirements for the project whenever required.

Thank You!